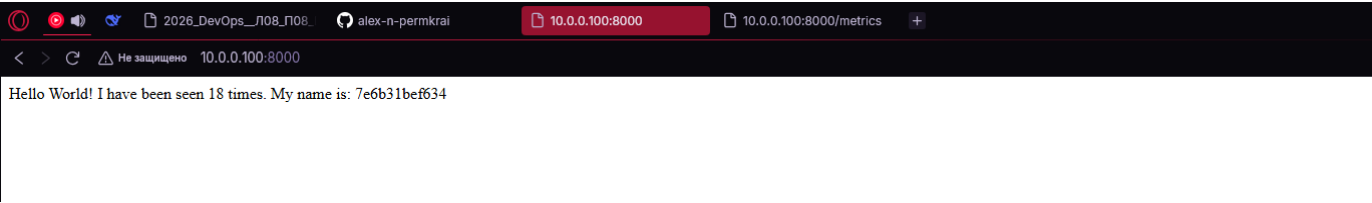
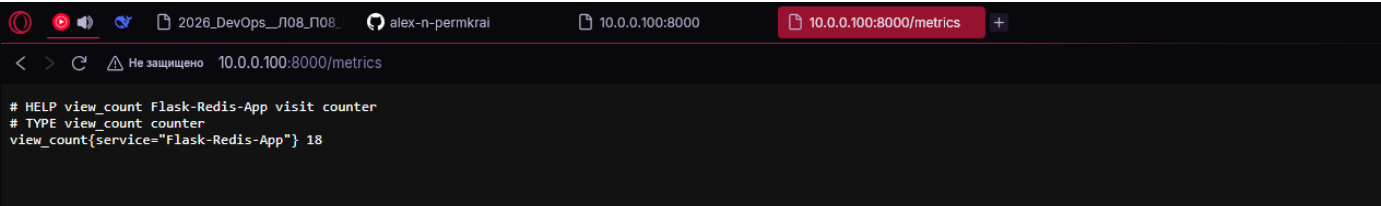


# Веб приложение flask+redis на 10.0.0.100



метрики для prometheus



Flask+Redis функционируют, порт 8000 доступен.

Endpoint /metrics отдаёт данные счётчика в формате Prometheus.

```
root@ubuntu-1:/home/user/devops9compose# curl http://localhost:8000
Hello World! I have been seen 20 times. My name is: 7e6b31bef634
root@ubuntu-1:/home/user/devops9compose# curl http://localhost:8000/metrics
# HELP view_count Flask-Redis-App visit counter
# TYPE view_count counter
view_count{service="Flask-Redis-App"} 20
root@ubuntu-1:/home/user/devops9compose# curl http://localhost:8000
Hello World! I have been seen 21 times. My name is: 7e6b31bef634
root@ubuntu-1:/home/user/devops9compose# curl http://localhost:8000/metrics
# HELP view_count Flask-Redis-App visit counter
# TYPE view_count counter
view_count{service="Flask-Redis-App"} 21
root@ubuntu-1:/home/user/devops9compose#
```

```
root@ubuntu-1:/home/user/devops9compose# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
7e6b31bef634   devops9compose-web  "flask run --debug"      4 minutes ago  Up 4 minutes    0.0.0.0:8000->5000/tcp, [::]:8000->5000/tcp  devops9compose-web-1
8b93a3318b24   redis:alpine    "docker-entrypoint.s..." 4 minutes ago  Up 4 minutes    6379/tcp                               devops9compose-redis-1
root@ubuntu-1:/home/user/devops9compose#
```

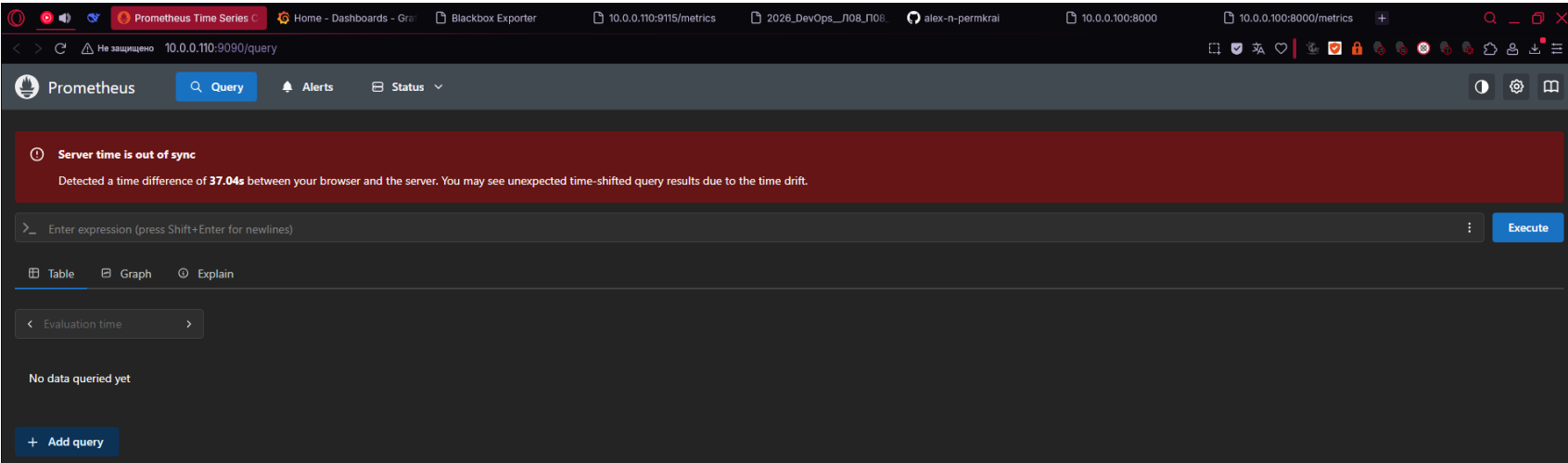
# Мониторинг (Prometheus + Grafana + Blackbox) на машине 10.0.0.110

## Запуск стека

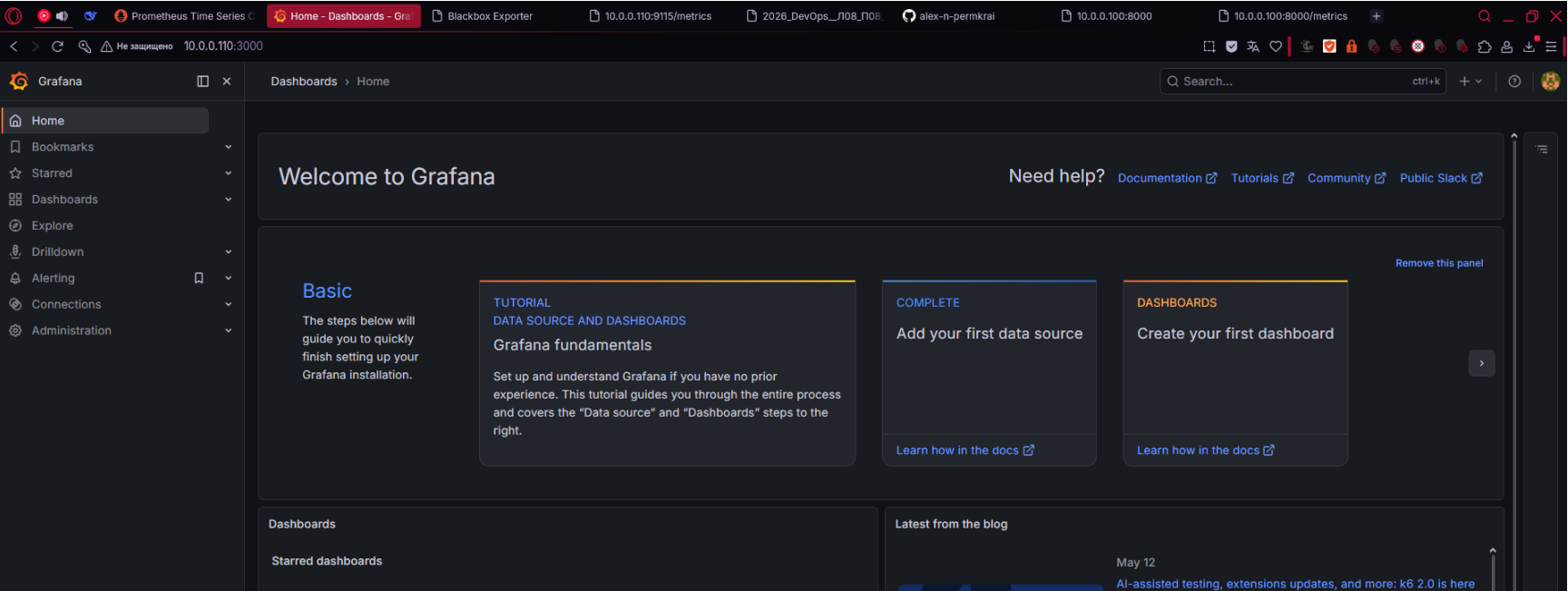
```
root@ubuntu-2:/home/user/devops9prom# docker compose up -d
[+] Running 27/27
  ✓ blackbox Pulled
    ✓ 0a17c6a98b51 Pull complete
    ✓ 788cf624215f Pull complete
    ✓ 48a18e2b648d Pull complete
    ✓ bfb096735d13 Pull complete
  ✓ prometheus Pulled
    ✓ cbbf09bbeab1 Pull complete
    ✓ 48cb70bc1631 Pull complete
    ✓ 92044327fe02 Pull complete
    ✓ 13199fe2e6d6 Pull complete
    ✓ 936791bae66e Pull complete
    ✓ 2acf9a40b14f Pull complete
    ✓ 61b07277e0e2 Pull complete
    ✓ cecfb7bc43f2 Pull complete
    ✓ 26cc3a207b02 Pull complete
    ✓ 921ec91b1376 Pull complete
  ✓ grafana Pulled
    ✓ b508b56b15dd Pull complete
    ✓ d4c060f3b932 Pull complete
    ✓ a3e1ec1858e3 Pull complete
    ✓ 4f4fb700ef54 Pull complete
    ✓ 38e67c1f799c Pull complete
    ✓ ae2bb398229f Pull complete
    ✓ 6513f998b3a1 Pull complete
    ✓ 74d8a3a79084 Pull complete
    ✓ 413bf89f4b8e Pull complete
    ✓ 01af952bd5a9 Pull complete
[+] Running 6/6
  ✓ Network devops9prom_default Created
  ✓ Volume devops9prom_grafana_data Created
  ✓ Volume devops9prom_prom_data Created
  ✓ Container grafana Started
  ✓ Container blackbox Started
  ✓ Container prometheus Started
root@ubuntu-2:/home/user/devops9prom#
```

## Проверка доступности

### Prometheus



Grafana



Blackbox

Blackbox Exporter

[Probe prometheus.io for http\\_2xx](#)  
[Debug probe prometheus.io for http\\_2xx](#)  
[Metrics](#)  
[Configuration](#)

Recent Probes

| Module   | Target                 | Result  | Debug                |
|----------|------------------------|---------|----------------------|
| http_2xx | http://10.0.0.100:8000 | Success | <a href="#">Logs</a> |
| http_2xx | https://student.psu.ru | Failure | <a href="#">Logs</a> |
| http_2xx | http://10.0.0.100:8000 | Success | <a href="#">Logs</a> |

Prometheus Time Series | Home - Dashboards - Grafana | Blackbox Exporter | 10.0.0.110:9115/metrics | 2026\_DevOps\_П08\_П08 | alex-n-permkrai | 10.0.0.100:8000 | 10.0.0.100:8000/metrics

Не защищено 10.0.0.110:9115/metrics

```
# HELP blackbox_exporter_build_info A metric with a constant '1' value labeled by version, revision, branch, goversion from which blackbox_exporter was built, and the goos and goarch for the build.
# TYPE blackbox_exporter_build_info gauge
blackbox_exporter_build_info{branch="HEAD",goarch="amd64",goos="linux",goversion="go1.25.5",revision="5a059bee8d8ffa4e75947c5055fb0abeefc582e6",tags="unknown",version="0.28.0"} 1
# HELP blackbox_exporter_config_last_reload_success_timestamp_seconds Timestamp of the last successful configuration reload.
# TYPE blackbox_exporter_config_last_reload_success_timestamp_seconds gauge
blackbox_exporter_config_last_reload_success_timestamp_seconds 1.7814263559481623e+09
# HELP blackbox_exporter_config_last_reload_successful Blackbox exporter config loaded successfully.
# TYPE blackbox_exporter_config_last_reload_successful gauge
blackbox_exporter_config_last_reload_successful 1
# HELP blackbox_module_unknown_total Count of unknown modules requested by probes
# TYPE blackbox_module_unknown_total counter
blackbox_module_unknown_total 0
# HELP go_gc_duration_seconds A summary of the wall-time pause (stop-the-world) duration in garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 1.7564e-05
go_gc_duration_seconds{quantile="0.25"} 5.4422e-05
go_gc_duration_seconds{quantile="0.5"} 8.1045e-05
go_gc_duration_seconds{quantile="0.75"} 0.000191408
go_gc_duration_seconds{quantile="1"} 0.000298473
go_gc_duration_seconds_sum 0.000713011
go_gc_duration_seconds_count 6
# HELP go_gc_gogc_percent Heap size target percentage configured by the user, otherwise 100. This value is set by the GOGC environment variable, and the runtime/debug.SetGCPercent function. Sourced from /gc/gogc:percent.
# TYPE go_gc_gogc_percent gauge
go_gc_gogc_percent 100
# HELP go_gc_gomemlimit_bytes Go runtime memory limit configured by the user, otherwise math.MaxInt64. This value is set by the GOMEMLIMIT environment variable, and the runtime/debug.SetMemoryLimit function. Sourced from /gc/gomemlimit:bytes.
# TYPE go_gc_gomemlimit_bytes gauge
go_gc_gomemlimit_bytes 9.223372036854776e+18
# HELP go_goroutines Number of goroutines that currently exist.
# TYPE go_goroutines gauge
go_goroutines 12
# HELP go_info Information about the Go environment.
# TYPE go_info gauge
go_info{version="go1.25.5"} 1
# HELP go_memstats_alloc_bytes Number of bytes allocated in heap and currently in use. Equals to /memory/classes/heap/objects:bytes.
# TYPE go_memstats_alloc_bytes gauge
go_memstats_alloc_bytes 6.971376e+06
# HELP go_memstats_alloc_bytes_total Total number of bytes allocated in heap until now, even if released already. Equals to /gc/heap/allocs:bytes.
# TYPE go_memstats_alloc_bytes_total counter
go_memstats_alloc_bytes_total 1.8817328e+07
# HELP go_memstats_buck_hash_sys_bytes Number of bytes used by the profiling bucket hash table. Equals to /memory/classes/profiling/buckets:bytes.
# TYPE go_memstats_buck_hash_sys_bytes gauge
go_memstats_buck_hash_sys_bytes 1.450709e+06
# HELP go_memstats_frees_total Total number of heap objects frees. Equals to /gc/heap/frees:objects + /gc/heap/tiny/allocs:objects.
# TYPE go_memstats_frees_total counter
go_memstats_frees_total 53572
# HELP go_memstats_gc_sys_bytes Number of bytes used for garbage collection system metadata. Equals to /memory/classes/metadata/other:bytes.
# TYPE go_memstats_gc_sys_bytes gauge
go_memstats_gc_sys_bytes 3.290896e+06
# HELP go_memstats_heap_alloc_bytes Number of heap bytes allocated and currently in use, same as go_memstats_alloc_bytes. Equals to /memory/classes/heap/objects:bytes.
# TYPE go_memstats_heap_alloc_bytes gauge
go_memstats_heap_alloc_bytes 6.971376e+06
# HELP go_memstats_heap_idle_bytes Number of heap bytes waiting to be used. Equals to /memory/classes/heap/released:bytes + /memory/classes/heap/free:bytes.
# TYPE go_memstats_heap_idle_bytes gauge
go_memstats_heap_idle_bytes 2.41664e+06
# HELP go_memstats_heap_inuse_bytes Number of heap bytes that are in use. Equals to /memory/classes/heap/objects:bytes + /memory/classes/heap/unused:bytes
# TYPE go_memstats_heap_inuse_bytes gauge
go_memstats_heap_inuse_bytes 9.281536e+06
# HELP go_memstats_heap_objects Number of currently allocated objects. Equals to /gc/heap/objects:objects.
# TYPE go_memstats_heap_objects gauge
go_memstats_heap_objects 9503
# HELP go_memstats_heap_released_bytes Number of heap bytes released to OS. Equals to /memory/classes/heap/released:bytes.
# TYPE go_memstats_heap_released_bytes gauge
```

Создание кастом дашборда grafana

